IDR305 21H Introduction to Economics for Sport Management Exam

Instructions (read carefully)

a. **3 hours and 30 minutes, including all practical and technical actions needed to upload in Inspera.**
b. **All supporting materials permitted. It is not allowed to cooperate with or receive help from others.**
c. **Justify all your answers with economic concepts, logical reasoning, and love for sport.**

Good Luck!

Question 1 (20 points)

Suppose there is a country that can produce two goods: X and Y. The country has 300 machines and 500 workers. Each machine can produce 4 units of X or 2 units of Y. Each worker can produce 1 unit of X and 3 units of Y.

a. Please draw a production possibilities frontier of this country and mark all the points (X and Y) where the curve changes its slope as well as the maximal production of X and Y. **(8 points)**
b. Suppose that the country needs to consume 1500 units of Y. in this case, how many units of X will it consume? **(3 points)**
c. Suppose there is a world trade, such that 1 unit of X is worth 4 units of Y. How many units of X and Y should the country produce and why? **(4 points)**
d. Suppose that during the world trade, the country still needs to consume the number of units of X that you found in question b. in this case, how many units of Y will it consume? **(5 points)**

Question 2 (20 points).

You are given the following functions:

1) Demand function: $Q^D = 25 - 2p$.
2) Supply function: $Q^S = 3p$

Where p is the price of a good in US dollars ($) and $Q^D and\ Q^S$ are the quantity demanded and supplied, respectively for that specific good.

a. What would be the price and the quantity in the equilibrium? **(4 points)**
b. Show on the graph (where appropriate) the cases where p=0, Q=0, and price and quantity in equilibrium. **(8 points)**
c. What is the producer surplus? **(3 points)**
d. What is consumer surplus? **(3 points)**
e. What is the total social welfare? **(2 points)**

Question 3 (20 points)

You are given the following firm's production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				50			
1	10							
2	18							
3	30							
4	45							

One unit of labor (L) costs 20$

a. Complete the table. **(10 points)**
b. What is the minimal price that the firm will produce in the short term? **(2 points)**
c. What is the minimal price that the firm will produce in the long term? **(2 points)**
d. It is known that each unit of production can be sold in the market for 240$ (p=240). What would be the production of the firm? **(2 points)**
e. What would be its profit in the short term? **(2 points)**
f. What would be its profit in the long term? **(2 points)**

Question 4 (10 points)

a. Show on a graph the effects of a price ceiling **(5 points)**
b. Provide a sport-related policy that embodies the concept of price ceiling. **(3 points)**
c. How are these two policies similar? How do they differ? **(2 points)**

Question 5 (10 points)

There is a market with 7 workers who should be allocated between two fields, given the following production function:

Field A

L	TP
0	0
1	7
2	13
3	18
4	22
5	23

Field B

L	TP
0	0
1	4
2	7
3	9
4	10
5	10

a. How many workers would work in each field if it is known that the price of one produced unit in field **A is 1$** and the price of one produced unit in field B is **3$**? **(6 points)**
b. What would be the wage in this market? **(4 points)**

Question 6 (5 points)

You are expected to receive a one-time sum of 100 NOK at the end of a five-years investment plan. What is the Present Value (PV) of this plan if the annual interest rate is 3% in the first two years and 5% per year after the first two years?

Question 7 (5 points)

You are expected to invest a one-time sum of 1000 NOK at the end of the sixth year from now and receive it at the end of the eighth year from now. What is the Future Value of this sum at the end of the eighth year from now if the annual interest rate is 3% in the first two years, 4% in the third year, 7% in the fourth year and 10% per year after that?

Question 8 (5 points)

Suppose that tickets for the final of the FIFA World Cup 2022 in QATAR are a substitute good for an organized 3-days trip in Tromsø to see the Northern Light. What would happen to demand for this 3-days trip if the price for the tickets to the final of the World Cup increases?

Question 9 (5 points)

Define the concept of “competitive balance” in sports economics and mention one possible way to promote it.